Marc Machaalani, MD

Education

Yale School of Medicine — New Haven, CT, USA

Traditional Internal Medicine Residency Program — 2026–Present

Dana-Farber Cancer Institute — Boston, MA, USA

Postdoctoral Research Fellowship — 2023–2026

Lebanese University, Faculty of Medical Sciences — Beirut, Lebanon

Doctor of Medicine (M.D.) — 2016–2023

Lebanese National Higher Conservatory of Music — Beirut, Lebanon

Bachelor's Degree in Piano Studies with High Distinction — 2006–2021

Honors and Awards

Merit Travel Grant — Feb 2026

OncLive National Fellows Forum for Genitourinary Cancers · San Francisco, CA, USA

Merit Award — Oct 2025

European Society of Medical Oncology (ESMO) Congress · Berlin, Germany

Conquer Cancer Merit Award — Jun 2025

American Society of Clinical Oncology (ASCO) Annual Meeting · Chicago, IL, USA

Conquer Cancer Merit Award — Feb 2025

American Society of Clinical Oncology Genitourinary (ASCO GU) Cancers Symposium · San Francisco, CA, USA

Merit Travel Grant — Feb 2025

OncLive National Fellows Forum for Genitourinary Cancers · San Francisco, CA, USA

Frank Shaughnessy Endowed Fellowship in Kidney Cancer — Dec 2024

Boston, MA, USA

Merit Travel Grant — Jan 2024

OncLive National Fellows Forum for Genitourinary Cancers · San Francisco, CA, USA

Merit Award for Distinguished Medical Students — Sep 2023

American Lebanese Medical Association (ALMA) · Beirut, Lebanon

Selected by the American Lebanese Medical Association (ALMA) as one of six recipients among 800+ medical students, based on the dean's recommendation.

Research Experience

Postdoctoral Research Fellow 2023–2026

- Medical Oncology Department at Dana-Farber Cancer Institute, Boston, MA
- Mentored by Toni K. Choueiri, MD (Professor) and Wenxin Xu, MD (Assistant Professor)
- Techniques: Western blotting, ELISA, multiplex immunoassays, protein quantification, DNA/RNA extraction, PCR, RT-qPCR, gel electrophoresis, plasmid cloning, bacterial transformation, Midi-Prep, dsDNA quantification, maintenance of mammalian cell lines, transfection, viability assays, DESeq2 analyses, GSEA (fgsea/msigdbr), regression models
- Focus on integrative analyses of circulating and tumor biomarkers in renal cell carcinoma, tumor–host interactions, immune evasion, the gut microbiome, and cancer genomics.

Research Volunteer 2020–2023

- Lebanese University, Faculty of Medical Sciences, Lebanon
- Published six peer-reviewed articles, including four as first author
- Mentored junior students in research projects

Ad Hoc Reviewer 2020–Present

- Reviewed 30+ manuscripts for 10+ journals including Nature
- Web of Science profile: https://www.webofscience.com/wos/author/record/AAU-9292-2021

Published Papers

1. Labaki C, Bakouny Z, Sanglier T, Saad E, Valero Martin M, Tarantino P, Schmidt AL, Shim J, Braun DA, Bouquet F, Joyce M, **Machaalani M**, Van Allen EM, Xie W, Tolaney SM, Choueiri TK. Novel framework for the validation of intermediate clinical endpoints in oncology real-world data. *J Natl Cancer Inst*. 2026 Oct 19:djag390. doi: 10.1093/jnci/djag390. PMID: 42898006.
2. Madsen KN, Kashima S, Soulati H, Wirth L, Ito K, Yochum ZA, Moritz V, Walker J, **Machaalani M**, Ghali F, Kenney P, Adeniran A, Hurwitz M, Choueiri TK, Braun DA. Pharmacologic HIF-2α inhibition preserves T cell effector function in clear cell renal cell carcinoma. *J Clin Invest*. 2026 Oct 6:e204613. doi: 10.1172/JCI204613. PMID: 42837196.
3. Dudipala H, Dugan AJ, Jariwala U, Ronski K, Mercer J, Saad E, El Masri J, **Machaalani M**, Saleh M, Viets Layng K, Yu Y, Signoretti S, Braun D, Pal SK, Choueiri TK, McKay RR. Molecular and immune correlates of lymphocyte activation gene-3 expression in renal cell carcinoma. *The Oncologist*. 2026;31(9):oyag317. doi: 10.1093/oncolo/oyag317. PMID: 42578740.
4. Suissa D, Fidelle M, Reich E, Pham TN, Thomas S, Björk JR, Liu P, Zhao L, Kitaoka K, Piard E, Lebhar I, Tian AL, Thelemaque C, Alves Costa Silva C, Deutsch E, Loriot Y, Segata N, Piccinno G, Hospers GAP, Maleki Vareki S, Silverman MS, Lenehan JG, Bataille V, Boulate D, Kuznetsova T, Weersma RK, Messaoudene M, Durand S, van der Aalst CM, de Koning HJ, Schuler-Thurner B, de Vries IJM, Rafie E, Saliby RM, **Machaalani M**, Haferkamp S, Schilling B, Porcari S, Ciccarese C, Iacovelli R, Cremolini C, Choueiri TK, Elkrief A, Kroemer G, Heinzerling L, Chamoto K, Ianiro G, Routy B, Derosa L, Paragios N, Zitvogel L. Metabolic determinants of cancer immunotherapy outcomes identified by plasma profiling. *Nature Medicine*. 2026. doi: 10.1038/s41591-026-04481-9. PMID: 42350644.
5. Saad E†, **Machaalani M**†, McDermott DF, Choueiri TK. The clinical landscape of HIF2α inhibitors in oncology. *Nature Reviews Clinical Oncology*. 2026. doi: 10.1038/s41571-026-01145-y. †Equal contribution. PMID: 42050152.

6. Roy AM, Bakouny Z, Dizman N, Paul MA, Ozay ZI, Lal BM, Nawfal R, Eid M, El Hajj Chehade R, **Machaalani M**, Butt A, Aggarwal C, Bekaii-Saab T, Chavez-MacGregor M, Lopes G, Marshall A, Murphy M, Owonikoko T, Xie W, Lee AJ, Choueiri TK. International Medical Graduates Representation at International Oncology Conference Meetings: An Analysis of ASCO Annual Meetings. *JCO Oncol Pract*. 2026 Apr;22(4):708-716. doi: 10.1200/OP-25-00067. Epub 2025 Jun 27. PMID: 40577682.
7. Nassar AH, Bou Farhat E, Abushukair H, Alchoueiry M, Salem S, **Machaalani M**, Adib E, Henske E, Babu P, Choueiri TK, Rakaee M, Busund LTR, Takabe Y, Lau SF, Rall K, Naqash AR, Jansen C, Wang X, Challa P, Tarantino P, Partridge AH, Tolaney SM, Kwiatkowski DJ, Winer EP. Molecular and clinical insights of trastuzumab deruxtecan efficacy in advanced breast cancer. *JNCI*. 2026 Apr 1;118(4):669-679. doi: 10.1093/jnci/djaf344. PMID: 41418409.
8. Yekedüz E, Bian W, Siegmund SE, **Machaalani M**, El Masri J, Saleh M, Saad E, Ascione L, El Halabi R, Steiner C, Mourad P, Berg SA, McGregor B, Mantia C, Ravi P, Xu W, Hirsch MS, Choueiri TK, Serzan M. Integrative Clinical and Molecular Evaluation of Renal Cell Carcinoma with Merlin Protein Deficiency and Biallelic Loss of NF2. *Clinical Cancer Research*. 2026. doi:10.1158/1078-0432.CCR-25-2871. PMID: 41524698.
9. Alves Costa Silva C†, **Machaalani M**†, Saliby RM†, Zhong C, Xie W, Pasolli E, Piccinno G, Dalban C, Fidelle M, Meurisse A, Vernerey D, Lee GS, Birebent R, Saad E, Steiner C, Flippot R, Barros-Monteiro J, Segata N, Thiery-Vuillemin A, Formenti S, Kuznetsova T, Escudier B, Derosa L, Zitvogel L, Albiges L. Soluble MAdCAM-1 as a biomarker in metastatic renal cell carcinoma. *Nature Medicine*. 2026. doi:10.1038/s41591-025-04067-x. †Equal contribution. PMID: 41501493.
10. Abu-Remaileh M, Stransky LA, Bhalerao N, Shirole NH, Jiang Q, Saad E, **Machaalani M**, Vigeant SM, Woldemichael H, Xu C, Lu J, Wei H, Liu Z, Sun W, Enomoto K, Choueiri TK, Pitarresi JR, Carr SA, Udeshi ND, Kaelin WG Jr. Targeting of HIF2-driven cachexia in kidney cancer. *Nat Med*. 2026 Jan;32(1):245-257. doi: 10.1038/s41591-025-04054-2. Epub 2025 Nov 28. Erratum in: *Nat Med*. 2026 Feb;32(2):766. doi: 10.1038/s41591-025-04054-2. PMID: 41315757; PMCID: PMC12823431.
11. Grubb T, Koochaki P, Matar S, Ghandour F, **Machaalani M**, Saad E, Tang C, Reznik E, Khatun J, Salem C, Dubasik N, Orlando DA, Guenther MG, Parajuli G, Srivastava RM, Martinez SR, Coker JA, Kotecha RR, Hakimi AA, Asara JM, Chan TA, Vanharanta S, Stauffer SR, Kaelin WG Jr, Signoretti S, Choueiri TK, Chakraborty AA. The SLC1A1/EAAT3 dicarboxylic amino acid transporter is an epigenetically dysregulated nutrient carrier that sustains oncogenic metabolic programs. *Nat Commun*. 2025;16(1):10012. doi:10.1038/s41467-025-64983-x. PMID: 41238532.
12. Dawidek MT, Posada Calderon L, Shah PK, Mittal R, Agridag Ucpinar B, Bakouny Z, Meli K, Saad E, Saliby RM, Woo HJ, Khandwala YS, Eid M, Labaki C, **Machaalani M**, Coleman J, Russo P, Kotecha RR, Carlo MI, Akin O, Braun DA, Choueiri TK, Reznik E, Chen YB, Motzer RJ, Voss MH, Hakimi AA. MLH1 mismatch repair deficiency predicts exceptional response to immune checkpoint inhibition in clear-cell renal cell carcinoma. *Eur Urol*. 2025 Nov 4:S0302-2838(25)04721-9. doi: 10.1016/j.eururo.2025.09.4162. Epub ahead of print. PMID: 41193371.
13. Sabarwal A, **Machaalani M**, Rawat L, Wedel J, Tabasum S, Sasamoto Y, Buerger F, Ascione J, Eid M, Semaan K, Saad E, Yang Y, Lee D, Hodi FS, Freedman ML, Lee GSM, Balan M, Choueiri TK, Pal S. Targeting AXL can effectively overcome c-Met-induced therapeutic resistance in renal cancer and promote tumor cell death through increased oxidative stress. *Cancer Lett*. 2025;633:217984. doi:10.1016/j.canlet.2025.217984. PMID: 40829740.

14. Labaki C, Saad E, Madsen KN, Hobeika C, Bi K, Alchoueiry M, Camp S, Hou Y, Bakouny Z, Matar S, El Ahmar N, Nyman J, Zhang L, Priolo C, Rout R, Daou M, Khabibullin D, Salem S, Schindler N, Saliby RM, Meli K, Wells C, Pimenta E, Takemura K, Marji J, Eid M, Sarlemi K, Xu J, Denize T, El Hajj Chehade R, **Machaalani M**, Nawfal R, Khattoun W, Saleh M, El Masri J, Haddad NR, Xu W, McGregor BA, Hirsch MS, Yilmaz R, Cheng DY, McDermott DF, Signoretti S, Romee R, Choueiri TK, Braun DA. Tumor-Intrinsic and Microenvironmental Determinants of Impaired Antitumor Immunity in Chromophobe Renal Cell Carcinoma. *J Clin Oncol*. 2025;43(23):2639-2654. doi:10.1200/JCO-25-00234. PMID: 40601888.
15. Yekeduz E, Zarba M, Saad E, El Hajj Chehade R, Eid M, Saliby RM, Steiner C, **Machaalani M**, Nawfal R, Semaan K, Ürün Y, Heng DYC, Choueiri TK. Comorbidity burden and effectiveness of immunotherapy in metastatic renal cell carcinoma. *Clin Genitourin Cancer*. 2025;23(4):102385. doi:10.1016/j.clgc.2025.102385. PMID: 40615325.
16. Fletcher K, **Machaalani M**, El Hajj Chehade R, Nassar AH, Nawfal R, Manos M, Menzies AM, Aboubakar-Nana F, Hasse JC, Pinato DJ, Johnson A, Olsson-Brown AC, Carlino MS, Malgieri A, Cortellini A, Singh A, Parikh K, Kim SY, Naqash AR, Long GV, Challa P, Choueiri TK, Sharon E, Shah S, Johnson DB. Immune checkpoint inhibitors for patients with preexisting autoimmune neurologic disorders. *JAMA Netw Open*. 2025;8(6):e2513727. doi:10.1001/jamanetworkopen.2025.13727. PMID: 40465294.
17. Steiner C†, **Machaalani M**†, Bonventre JV, McDermott DF, Choueiri TK, Xu W. KIM-1 as a prognostic marker in renal cell carcinoma. *Eur Urol Focus*. 2025;11(3):432-435. doi:10.1016/j.euf.2025.06.012. †Equal contribution. PMID: 40619332.
18. Yadav M, Sharma A, Patne K, Tabasum S, Suryavanshi J, Rawat L, **Machaalani M**, Eid M, Singh RP, Choueiri TK, Pal S, Sabarwal A. AXL signaling in cancer: from molecular insights to targeted therapies. *Signal Transduct Target Ther*. 2025;10(1):37. doi:10.1038/s41392-024-02121-7. PMID: 39924521.
19. Xu W, Birch G, Melki A, Moritz V, Bharadwaj M, Schindler NR, Labaki C, Saliby RM, Dinh K, Horst JT, Sun M, Kashima S, Hugaboom M, Dighe A, **Machaalani M**, Lee GS-M, Hurwitz M, McGregor BA, Hirsch MS, Shukla SA, McDermott DF, Signoretti S, Romee R, Choueiri TK, Braun DA. Progressive natural killer cell dysfunction in advanced-stage clear-cell renal cell carcinoma and association with clinical outcomes. *ESMO Open*. 2025;10(2):104105. doi:10.1016/j.esmoop.2024.104105. PMID: 39813824.
20. **Machaalani M**, Eid M, Semaan K, El Hajj Chehade R, Nawfal R, Baca SC, Choueiri TK. Liquid biopsy in renal cell carcinoma. *Oncologist*. 2024;29(10):821-823. doi:10.1093/oncolo/oyae230. PMID: 39187381.
21. El Zarif T†, **Machaalani M**†, Nawfal R†, Nassar AH, Xie W, Choueiri TK, Pomerantz M. TERT Promoter Mutations Frequency Across Race, Sex, and Cancer Type. *Oncologist*. 2024;29(1):8-14. doi:10.1093/oncolo/oyad208. †Equal contribution PMID: 37462445.
22. **Machaalani M**, El Masri J, El Ayoubi LM, Matar B. Cancer research activity in the Arab world: a 15-year bibliometric analysis. *J Egypt Public Health Assoc*. 2022;97(1):26. doi:10.1186/s42506-022-00120-6. PMID: 36385361.
23. **Machaalani M**, Fakhry B, Zwaideh M, Mendelek K, Mahmoud N, Hammoud T, Chahine MN. Knowledge, Attitude, and Practice Toward Cardiovascular Diseases in the Lebanese Population. *Global Heart*. 2022;17(1):47. doi:10.5334/gh.1138. PMID: 36051313.
24. **Machaalani M**, Seifeddine H, Ali A, Bitar H, Briman O, Chahine MN. Knowledge, Attitude, and Practice Toward Hypertension Among Hypertensive Patients Residing in Lebanon. *Vasc Health Risk Manag*. 2022;18:541-553. doi:10.2147/VHRM.S367187. PMID: 35860751.
25. El Zarif T, Yibrin M, De Oliveira-Gomes D, **Machaalani M**, Nawfal R, Bittar G, Bahmad HF, Bitar N. Overcoming Therapy Resistance in Colon Cancer by Drug Repurposing. *Cancers (Basel)*. 2022;14(9):2105. doi:10.3390/cancers14092105. PMID: 35565237.
26. El Ayoubi LM, El Masri J, **Machaalani M**, El Hage S, Salameh P. Contribution of Arab World in Transplant Research: A PubMed-Based Bibliometric Analysis. *Transpl Immunol*. 2021;68:101432. doi:10.1016/j.trim.2021.101432. PMID: 34186171.

Oral Presentations (First Author)

1. **Machaalani M**, Alves Costa Silva C, Saliby RM, Zhong C, Xie W, Pasolli E, Piccinno G, Dalban C, Fidelle M, Lee GSM, Birebent R, Saad E, Steiner C, Segata N, Thiery-Vuillemin A, Escudier B, Derosa L, Zitvogel L, Albiges L, Choueiri TK. Soluble MAdCAM-1 predicts outcomes in patients with metastatic renal cell carcinoma: results from three independent clinical trials. ESMO Congress 2025, Berlin, Germany (Oct 2025). 10.1016/j.annonc.2025.08.3215
2. **Machaalani M**, Saliby RM, Zhong C, Saad E, Steiner C, Yekeduz E, Ascione L, Saleh M, El Masri J, Barrios P, Liu X, Cai T, Sun M, Lee GSM, Xie W, Signoretti S, McDermott D, Choueiri TK, Xu W. Integrative clinical, genomic, transcriptomic, and immunopathologic characterization of circulating KIM-1. Kidney Cancer Research Summit (KCRS) 2025, Boston, MA, USA (Jul 2025).
3. **Machaalani M** et al. Implementing Biomarkers of Immunotherapy for Genitourinary Cancers. SITC Leveraging Biomarkers to Accelerate IO Drug Development Workshop, Boston, MA, USA (May 2025).
4. **Machaalani M**, Saliby RM, Zhong C, Saad E, Steiner C, Liu X, Cai T, Sun M, Lee GSM, Xie W, McDermott DF, Choueiri TK, Xu W. KIM-1 as a circulating biomarker in metastatic RCC: Post-hoc analysis of JAVELIN Renal 101. OncLive National Fellows Forum: Genitourinary Cancers 2025, San Francisco, CA, USA (Feb 2025).
5. **Machaalani M**, Saad E, Labaki C, Song L, Pan W, Saliby RM, Bakouny Z, Wall M, Semaan K, Eid M, Lee GSM, Braun DA, Liu XS, Choueiri TK. Host immune signatures as predictors of response to immunotherapy-based regimens in metastatic renal cell carcinoma. OncLive National Fellows Forum: Genitourinary Cancers 2024, San Francisco, CA, USA (Jan 2024).
6. **Machaalani M**, Fakhry B, Farhat K, Alhaddad J, Rahmeh Y, Ghiya P, Awad DC, Zaiter A, Louka JG, Olaywan L, Halawi A, Cherry H, Ghazal M, Sahili M, Atallah B, Naja W, Chammas E, Asmar R, Yared N, Chahine MN. Perspectives on knowledge, precautionary behaviors, and psychological status of patients with cardiovascular diseases during the first wave of the COVID-19 pandemic in Lebanon. Demystifying COVID-19: Lessons Learned and Future Opportunities conference, Beirut, Lebanon (Jul 2021).

Poster Presentations (First Author)

1. **Marc Machaalani**, Renee Maria Saliby, Carolina Alves Costa Silva, Eddy Saad, Clara Steiner, Caiwei Zhong, Emre Yekeduz, Xiaowen Liu, Liliana Ascione, Pablo Barrios, Ti Cai, Gwo-Shu Mary Lee, Wanling Xie, Sabina Signoretti, Maxine Sun, Laurence Zitvogel, Laurence Albiges, David F. McDermott, Wenxin Xu, and Toni K. Choueiri. A multi-axis biomarker model integrating KIM-1 and MAdCAM-1 in metastatic renal cell carcinoma. ASCO Annual Meeting 2026, Chicago, IL, USA (Jun 2026). 10.1200/JCO.2026.44.16_suppl.4554
2. **Marc Machaalani**, Eddy Saad, Chris Labaki, Ziad Bakouny, Carsten Schroeder, Yutonh Liu, Fanny Bouquet, Maureen Joyce, Bradley Alexander McGregor, Joaquim Bellmunt, Wanling Xie, and Toni K. Choueiri. Real-world endpoints as surrogates for overall survival in metastatic bladder cancer. ASCO Genitourinary Cancers Symposium 2026, San Francisco, CA, USA (Feb 2026). 10.1200/JCO.2026.44.7_suppl.693
3. **Marc Machaalani**, Amin H. Nassar, Razane El Hajj Chehade, Clara Steiner, Michel Alchoueiry, Marc Eid, Samer Salem, Emre Yekedüz, Renee Maria Saliby, Caroline S Jansen, Pavan Challa, Xiao Wang, Elias Bou Farhat, Wadih Issa, Hassan Kawtharany, Hamzah Abu-Sbeih, Mitali Shah, Ayesha Aijaz, Sagal Pannu, Paolo Tarantino, Arjun Mittra, Jun Zhang, Tian Zhang, Elizabeth P Henske, Sara M Tolaney, Sarah E Schellhorn, Toni K Choueiri, Abdul Rafeh Naqash. Pan-tumor multicenter study of the impact of prior immune-related adverse events (irAEs) on safety and efficacy of subsequent antibody-drug conjugate (ADC). Society for Immunotherapy of Cancer (SITC) annual meeting 2025, National Harbor, MD, USA (Sep 2025). https://jitc.bmj.com/content/13/Suppl_2/A454

4. **Machaalani M**, Saliby RM, Saad E, Steiner C, Zhong C, Yekeduz E, Liu X, Eid M, El Hajj Chehade R, Ascione L, El Masri J, Saleh MJ, Cai T, Sun M, Lee GSM, Xie W, Signoretti S, McDermott DF, Choueiri TK, Xu W. Integrative clinical, genomic, and transcriptomic characterization of circulating KIM-1 in metastatic RCC. ASCO Annual Meeting 2025, Chicago, IL, USA (May 2025). 10.1200/JCO.2025.43.16_suppl.4546
5. **Machaalani M**, Saliby RM, Zhong C, Saad E, Steiner C, Liu X, Cai T, Sun M, Lee GSM, Xie W, McDermott DF, Choueiri TK, Xu W. KIM-1 as a circulating biomarker in metastatic RCC: Post-hoc analysis of JAVELIN Renal 101. ASCO Genitourinary Cancers Symposium 2025, San Francisco, CA, USA (Feb 2025). 10.1200/JCO.2025.43.5_suppl.594
6. **Machaalani M**, Chehade REH, Nassar A, Zarif TE, Saad E, Yekeduz E, Saliby RM, Steiner C, Nawfal R, Semaan K, Eid M, Lee GSM, Pomerantz M, McKay RR, Tarantino P, Braun DA, Sun M, Tolaney SM, Choueiri TK. ERBB2 amplifications across sex, race, and cancer types. ESMO Congress 2024, Barcelona, Spain (Sep 2024). 10.1016/j.annonc.2024.08.2181
7. **Machaalani M**, Sabarwal A, Rawat L, Balan M, Eid M, Saad E, Semaan K, Yang Y, Ascione J, Freedman M, Lee GS, Pal S, Choueiri TK. Inhibition of AXL along with c-Met overcomes resistance development in renal cell carcinoma. Kidney Cancer Research Symposium 2024, Boston, MA, USA (Jul 2024).
8. **Machaalani M**, El Zarif T, Stockhammer P, Semaan K, Eid M, El Hajj Chehade R, Nawfal R, Yekeduz E, Nassar A, Lee GSM, Baca S, Choueiri TK, Grant MJ, Bellmunt J. ERBB2 mutations and association with molecular phenotype in urothelial carcinoma. ASCO Annual Meeting 2024, Chicago, IL, USA (May 2024).

Leadership and Extra-Curricular Activities

KidneyCAN Advocacy 2024–2026
Advocate for kidney cancer research funding through the Kidney Cancer Research Program (KCRP) by meeting with U.S. Representatives and Senators.

Lank Center for Genitourinary Oncology Journal Club 2023–2026
Organize and lead monthly journal clubs on GU oncology for faculty and research fellows.

Health Outreach to the Middle East (HOME) 2022–2023
Participated in mobile clinics providing free healthcare services to underserved communities.

Lebanese Cardiology Group 2021–2023
Oversaw a nationwide research project on CPR knowledge and sudden cardiac arrest (SCA) awareness involving 2,200+ participants.

Lebanese Oncology Interest Group 2021–2023
Led workshops, volunteering opportunities, fundraising, and free cancer clinics for underserved patients.

Lebanese Association of Medical Services for Addictions 2021–2023
Organized public education campaigns to reduce stigma around addiction and mental health; participated in community outreach and medical education programs.

Medicine Chess Club 2018–2021
Founded and led the chess club; organized training sessions; won multiple chess tournaments including the 2019 Lebanese University Championship.

Lebanese National Higher Conservatory of Music 2006–2021
Completed formal training in solfège, music theory, harmony, and counterpoint; tenor in Youth Choir and accompanist for violin rehearsals and performances.

Languages

- **Arabic:** native language
- **English and French:** full professional proficiency
- **German:** limited working proficiency